PROJECT DOCUMENTATION

Skill-Based Job Matching & Career Guidance Portal

Domain: Web Design & User Interface Engineering

Software Hackathon 2026

Day 1 & Day 2 — UI/UX Design Plan

Prepared: 31 July 2026

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GITHUB LINK: github.com/Nithish-K-63/Techtide
DEPLOYMENT LINK: https://techtide-gamma.vercel.app

1. Project Overview

The Skill-Based Job Matching & Career Guidance Portal is a web application designed to help job seekers discover roles that match their existing skill sets, identify skill gaps, and receive guided career advice. This documentation covers the UI/UX design process undertaken during the Software Hackathon 2026, spanning two days of structured design work — from research and design-thinking through to a fully interactive prototype and final presentation.

1.1 Project Title

Title: Skill-Based Job Matching & Career Guidance Portal

Domain: Web Design & User Interface Engineering

Event: Software Hackathon 2026

Scope: Day 1 & Day 2 — UI/UX Design Plan

1.2 Objective

To design an intuitive, accessible, and visually consistent web platform that connects users' skills to relevant job opportunities and career guidance resources, following an end-to-end design process: research, information architecture, visual design, prototyping, usability testing, and final presentation.

1.3 Project Description

The Skill-Based Job Matching & Career Guidance Portal is a web platform that connects a user's existing skill set directly to relevant job opportunities, rather than relying on generic keyword search alone. Users create a profile by entering or importing their skills, and the platform analyzes those skills against real job requirements to surface a ranked list of matches. Alongside matching, the portal identifies gaps between a user's current skills and the skills a target role demands, and points them toward focused learning resources to close those gaps. A dedicated careerguidance layer offers curated articles, learning paths, and mentorship prompts, making the portal not just a job board but an ongoing career-growth companion.

1.4 Proposed Solution

Most existing platforms solve only one part of the job-search problem — either listings, or networking, or generic learning content — leaving users to stitch the pieces together themselves. The proposed solution brings three capabilities into a single, connected experience:

- Skill-first matching — jobs are ranked by how well a user's actual skills fit the role, with a transparent match score
- Visual skill-gap analysis — a clear breakdown of which skills a user already has and which ones are missing for a target role
- Guided upskilling — each identified gap links directly to a relevant course, article, or resource, closing the loop between finding a gap and fixing it

This turns the portal into an end-to-end loop — discover a match, understand the gap, close the gap, re-apply with a stronger profile — instead of a one-time listing search.

1.5 Comparison with Existing Platforms

A review of the platforms job seekers rely on today — LinkedIn, Naukri, and Indeed — shows a consistent gap: none of them combine skill-gap analysis with guided learning inside the matching flow itself.

Attribute	LinkedIn	Naukri	Indeed
Core Focus	Networking + job search	Resume database for recruiters	Job listing aggregator
Matching Style	Profile & activity based	Recruiter keyword search	Keyword + location search
Skill Gap Analysis	Not present	Not present	Not present
Career Guidance	Generic articles	None	None
Networking	Strong	None	None
Best For	MNCs, senior roles	Indian IT, high volume	Entry-level, broad reach

Unique Differentiators of This Portal

- Skill-gap visualization directly on the match screen — a capability none of the three platforms above offer
- Match score shown with reasoning, not just a percentage, so users understand why a role fits
- Learning resources linked directly to each identified skill gap, closing the loop within one platform
- Dedicated focus on career switchers, a segment underserved by existing job boards
- A distraction-free, single-purpose interface — no networking feed or unrelated content competing for attention

1.6 Repository

Project Repository: github.com/Nithish-K-63/Techtide

The application code for this project is maintained in the linked repository. Refer to the repository's README and commit history for implementation-level detail alongside the design documentation in this report.

2. Design Process Timeline

The design work is organized into six phases across two days, as summarized below.

2.1 Day 1 — Foundation & Prototype

Phase	Objective	Expected Output
Phase 1	**User Research & Design Thinking**	*Understand the problem and design the best user experience.* • Analyze Problem Statement • Identify Target Users • User Personas • User Journey Mapping • Competitor Analysis • Define Design Goals

Phase 2	**UI/UX Planning & Design System**	*Create the application's visual foundation before designing screens.* • Information Architecture • Sitemap • Design System
Phase	**Objective**	**Expected Output**
		• Color Palette • Typography • Components • Grid System • Design Tokens
Phase 3	**UI Design & Interactive Prototype**	*Design all application screens and user interactions.* • Login Screens • Dashboard Design • Forms • Mobile Responsive Design • Auto Layout • Interactive Prototype • Design Validation

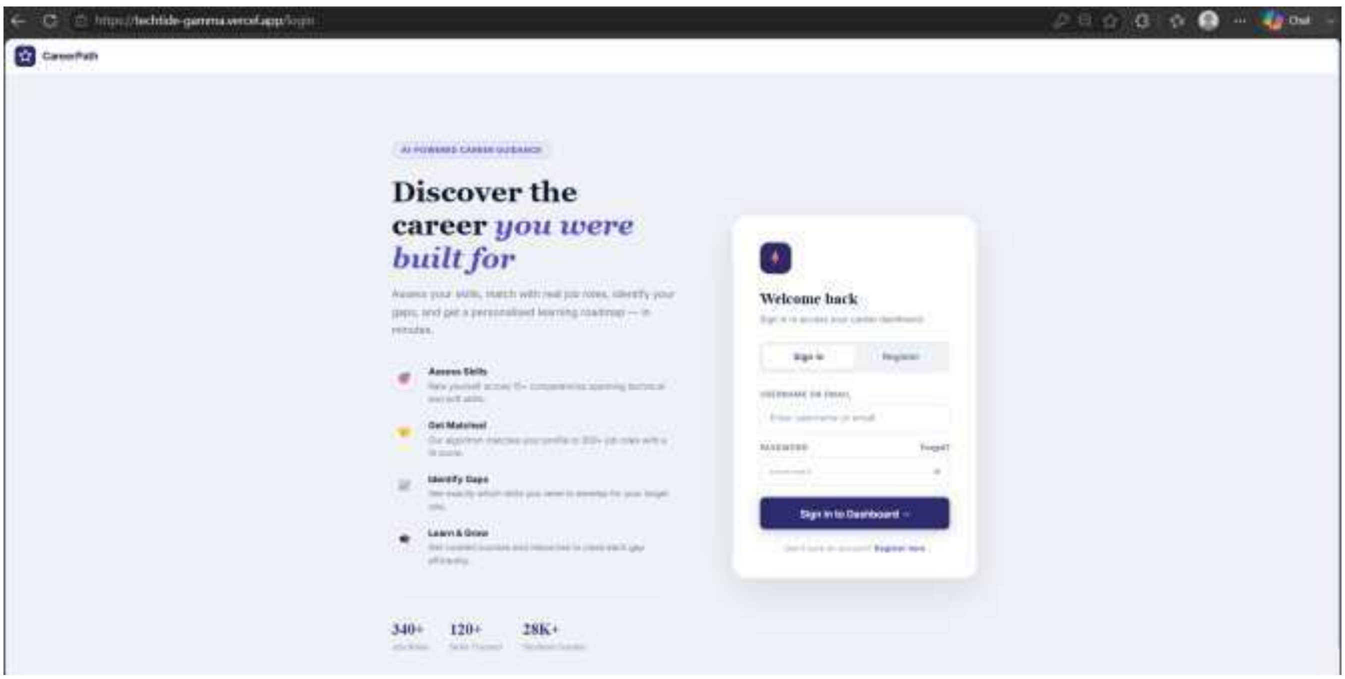
2.2 Day 2 — Refinement & Delivery

Phase	Objective	Expected Output
Phase 4	**High Fidelity Design**	*Complete all remaining application screens.* • Complete UI Screens • Icons • Illustrations • Empty States • Error Screens • Loading Screens
Phase 5	**Usability Testing & Iteration**	*Improve the design based on usability feedback.* • Design Review • Accessibility Check • User Testing • Design Improvements • Developer Handoff Preparation
Phase 6	**Final Presentation**	*Present the complete UI/UX solution.* • Design Process • User Flow • Design System • Interactive Prototype Demo • Challenges • Future Improvements

3. Phase Details

Phase 1 — User Research & Design Thinking

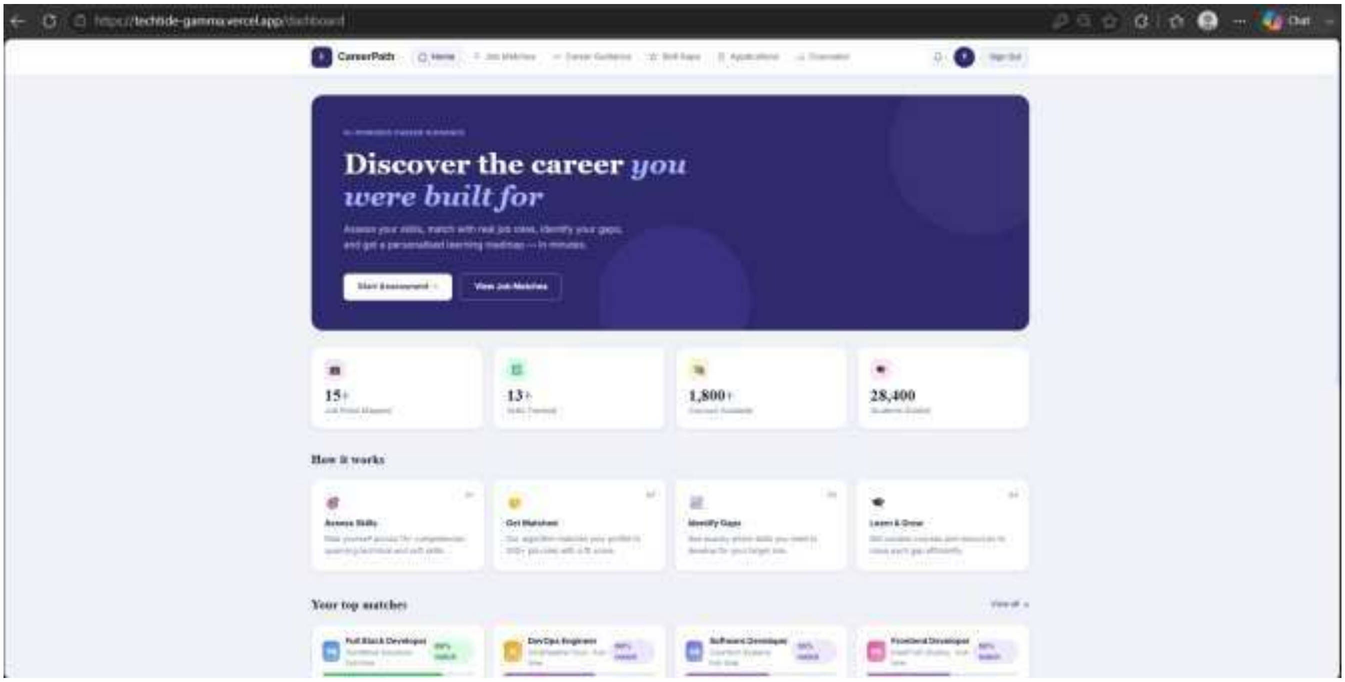
This phase focuses on building a clear understanding of the problem space and the needs of the target audience before any visual design begins.

- Analyze the problem statement to define the core challenge the portal solves
- Identify target users — job seekers, students, and career switchers
- Develop user personas representing key user segments
- Map the user journey from onboarding to job/career recommendations
- Conduct a competitor analysis of existing job-matching platforms • Define clear, measurable design goals
- 

Phase 2 — UI/UX Planning & Design System

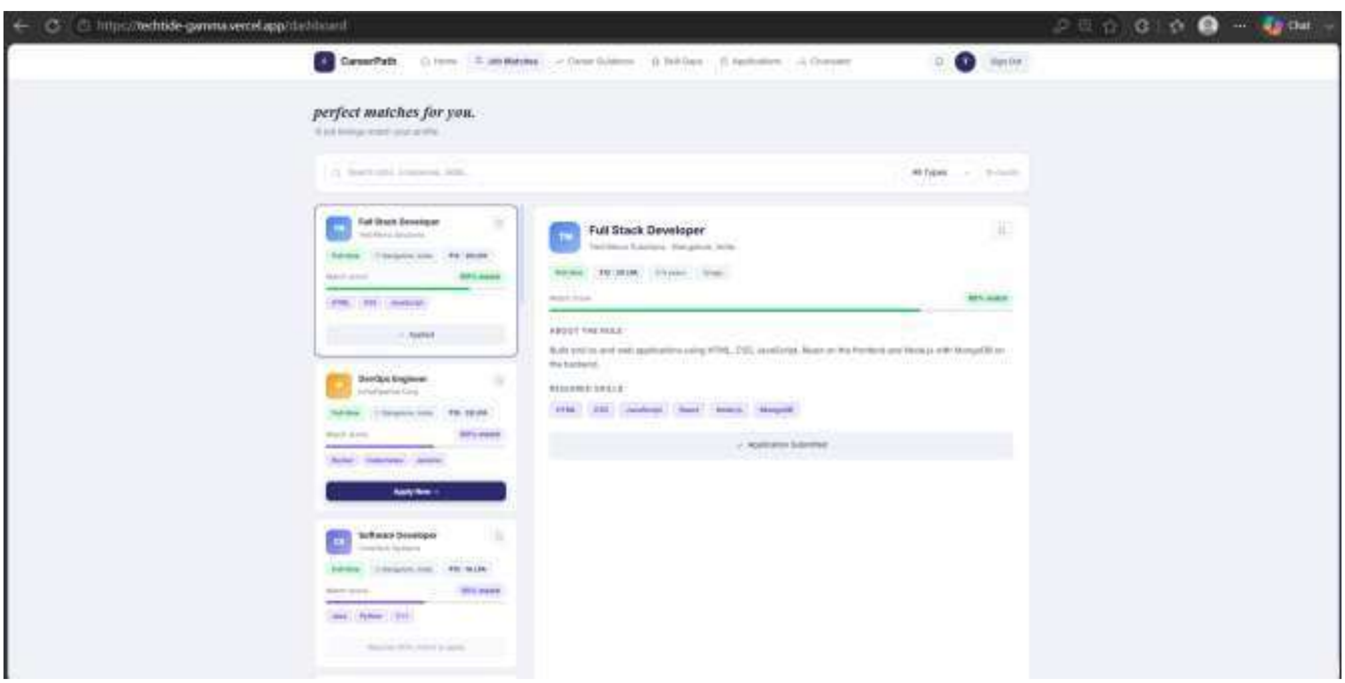
This phase establishes the visual foundation and structural blueprint of the application.

- Define the information architecture of the platform
- Create a sitemap outlining all pages and their relationships
- Build a reusable design system
- Select a cohesive color palette
- Establish a typography scale
- Design core UI components
- Set up a responsive grid system
- Define design tokens for consistency across screens

- 

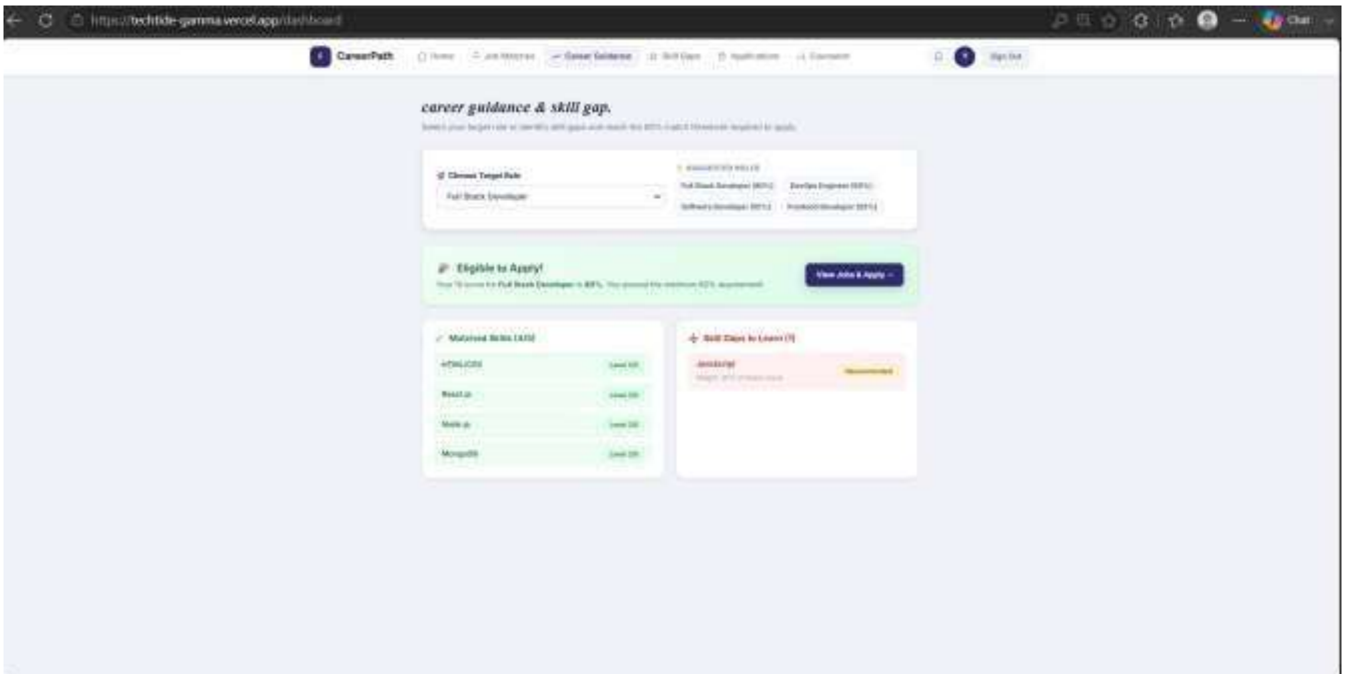
Phase 3 — UI Design & Interactive Prototype

This phase translates the design system into actual application screens and interactions.

- Design login and authentication screens
- Design the main dashboard
- Create forms for skills input, job preferences, and profile setup
- Ensure mobile responsive design across breakpoints
- Apply auto layout for scalable, flexible components
- Build an interactive prototype linking key user flows
- Perform design validation against defined goals
- 

Phase 4 — High Fidelity Design

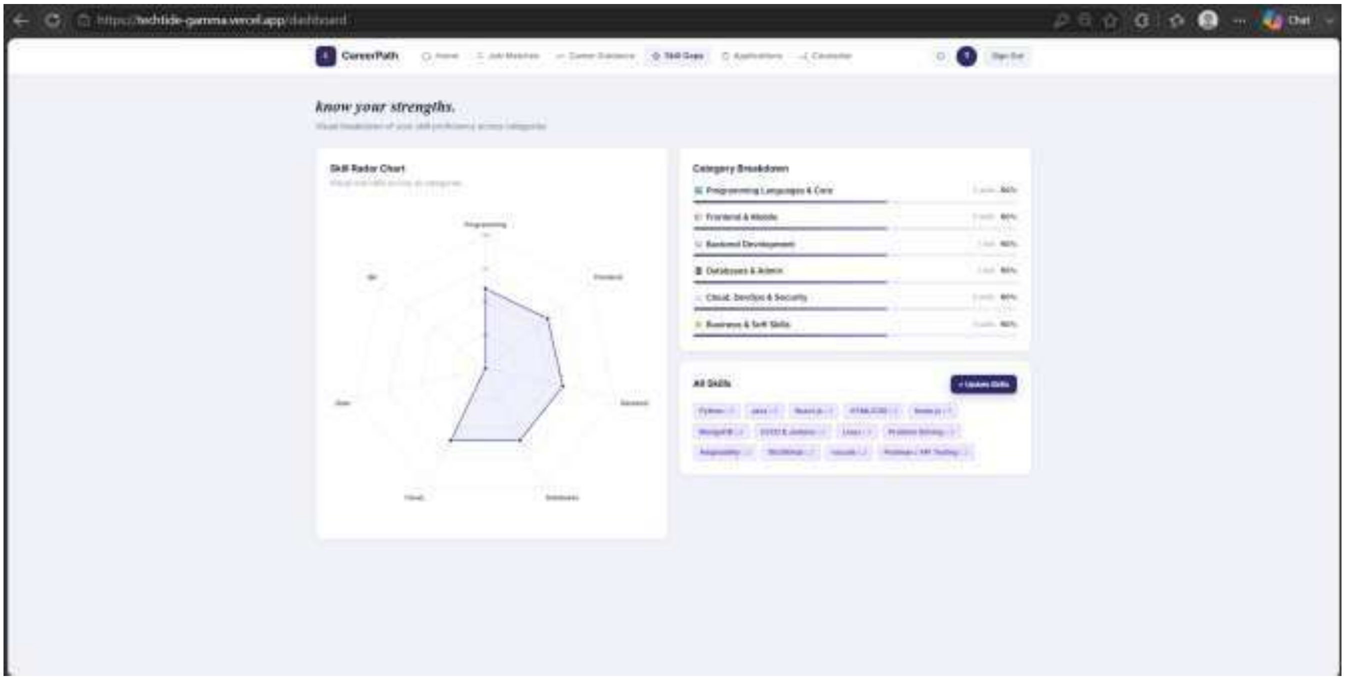
This phase completes the remaining screens with polished, production-ready visuals.

- Complete all remaining UI screens
- Design custom icons
- Create supporting illustrations
- Design empty states for zero-data scenarios
- Design error screens for failure states
- Design loading screens for wait states
- 

Phase 5 — Usability Testing & Iteration

This phase validates the design with real feedback and prepares it for development handoff.

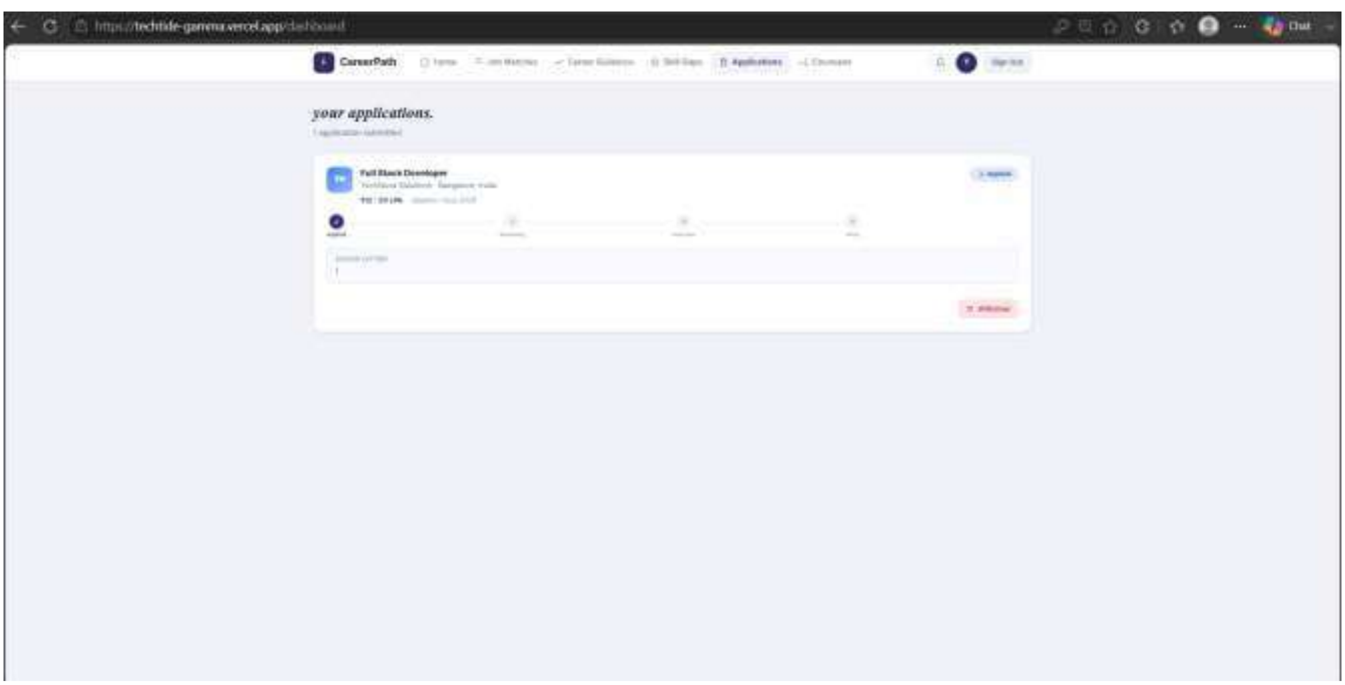
- Conduct internal design reviews
- Perform an accessibility check against WCAG guidelines
- Run user testing sessions to gather feedback
- Apply design improvements based on findings
- Prepare developer handoff documentation and assets

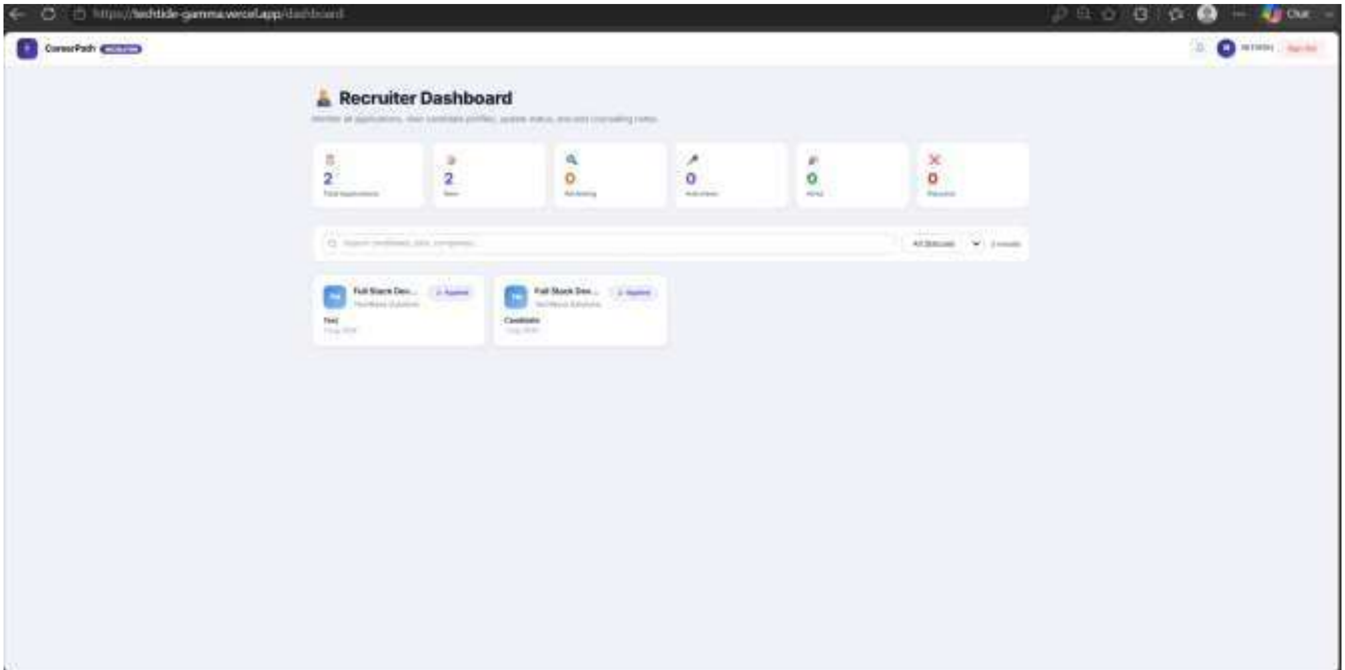
- 

Phase 6 — Final Presentation

This phase communicates the complete UI/UX solution to stakeholders and judges.

- Present the overall design process
- Walk through the core user flow
- Showcase the design system
- Demo the interactive prototype
- Discuss challenges faced during the project
- Outline future improvements and next steps

- 

- 

4. Usability Testing Summary

Phase 5 usability testing was conducted with a small group of representative users to validate the high-fidelity design before final handoff.

4.1 Method

Participants completed three core tasks — onboarding, reviewing job matches, and exploring a skill gap — while thinking aloud, followed by a short feedback interview.

4.2 Key Findings

- Onboarding flow was rated clear, though skill-tag suggestions needed more visibility
- Match percentage on job cards was well understood and appreciated
- Skill gap screen needed clearer next-step calls to action

4.3 Resulting Improvements

- Increased visual prominence of suggested skill tags during onboarding
- Added a clear 'Start Learning' call to action on each skill gap
- Simplified filter controls on the job matches screen

5. Challenges & Future Improvements

5.1 Challenges Faced

- Balancing information density on the dashboard without overwhelming new users
- Designing a skill-matching visualization that felt intuitive at a glance

- Fitting the full onboarding flow comfortably within mobile breakpoints

5.2 Future Improvements

- Introduce AI-assisted skill suggestions based on resume parsing
- Add a mentor-matching module connected to the guidance section
- Expand accessibility testing to include assistive-technology users
- Explore a progress-tracking view for in-progress learning paths

6. Phase-Wise Work Summary

The table below consolidates the work completed and features delivered in each of the six phases, from initial research through to the final presentation.

Phase	Work Done	Key Features Delivered
Phase 1	Research completed on target users, competitors, and problem definition.	User personas, journey map, competitor analysis, design goals
Phase 2	Visual foundation and structure planned before screen design began.	Sitemap, color palette, typography, components, design tokens
Phase 3	Core screens designed and connected into a clickable prototype.	Login, dashboard, forms, responsive layout, interactive prototype
Phase 4	Remaining screens completed with productionready visuals.	Icons, illustrations, empty/error/loading states
Phase	Work Done	Key Features Delivered
Phase 5	Design tested with users and refined based on feedback.	Accessibility check, usability findings, developer handoff assets
Phase 6	Complete solution consolidated and presented.	Design process walkthrough, prototype demo, challenges & next steps

7. Conclusion

This structured, two-day design process ensures that the Skill-Based Job Matching & Career Guidance Portal is built on solid user research, a consistent design system, and validated usability — resulting in a polished, presentationready UI/UX solution by the end of the hackathon. The resulting design system, screen set, and handoff documentation give the development team a clear, well-tested foundation to build on beyond the hackathon.